ALGEBRA 1 · UNIT 7: EXPONENTS AND SCIENTIFIC NOTATION

Scale of the Universe

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students explore the very large and very small with scientific notation: comparing sizes from viruses to planets, timing light across the solar system, and designing a scale model of space that fits in the school hallway.

Exponent rules become tools for comparing quantities ('how many times bigger?') and computing rates. Scaling the solar system to a hallway makes the size of space memorable and requires careful division of powers of ten.

Standards Addressed

- 8.EE.3–4 Scientific notation (applied)
- N-RN.1–2 Properties of exponents (Honors: rational exponents preview)
- A-SSE.2 Use structure to rewrite expressions
- N-Q.1 Use units to understand problems

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with the hero image: Is this picture to scale? (No!) Part 1 notation and comparisons.
Day 2	Part 2: travel times with the speed of light and spacecraft.
Day 3	Part 3: build the scale model plan; optionally walk it out in the hallway.

Materials

- Student handout and data cards
- Calculators
- Meter sticks or measuring tape (Day 3)
- Optional: a powers-of-ten video

How the Three Versions Differ

Level	What changes
CP2	Converting between forms, ordering, and dividing numbers with the same coefficient; a guided scale model.
CP1	Multiplying and dividing in scientific notation with normalization, light-years, and a two-planet scale model.
Honors	Powers and negative exponents, a log-scale number line, error analysis, and a model with two different scales.

Watch For (Common Misconceptions)

- Adding exponents when dividing.
- Leaving answers like 24×10^5 (not in scientific notation).
- Moving the decimal the wrong way for negative exponents.

Extension Ideas

- Walk the scale model in the hallway and take photos.
- Research distances to other stars.
- Compare the number of cells in the body to stars in the galaxy.

Answer Key – CP2

Part 1: Big & Small

#	Task	Answer
1	Write the Earth-to-Moon distance in standard form. Then write 56,000,000 km in scientific notation.	384,000 km; 5.6×10^7 km
2	Write the width of a red blood cell (8×10^{-6} m) in standard form.	0.000008 m
3	Order these from smallest to largest: human hair, flu virus, red blood cell.	Flu virus (1×10^{-7}), red blood cell (8×10^{-6}), human hair (8×10^{-5})
4	How many times wider is a human hair than a red blood cell? Show the division.	$1 \times 10^1 = 10$ times wider

Part 2: Speed of Light

#	Task	Answer
5	How many seconds does it take light from the Sun to reach Earth? How many minutes is that?	$(1.5 \times 10^8) \div (3 \times 10^5) = 0.5 \times 10^3 = 500$ seconds ≈ 8.3 minutes
6	How many seconds does it take light to travel from Earth to the Moon?	$(3.84 \times 10^5) \div (3 \times 10^5) = 1.28$ seconds
7	A spacecraft travels 4×10^4 km per hour. How many hours does it take to reach the Moon?	$(3.84 \times 10^5) \div (4 \times 10^4) = 0.96 \times 10^1 = 9.6$ hours
8	About how many times heavier is Earth than the Moon?	$(5.97 \times 10^{24}) \div (7.35 \times 10^{22}) \approx 0.81 \times 10^2 \approx 81$ times

Part 3: Design a Hallway Model

#	Task	Answer
9	How far from the Sun (in meters) should Earth be placed in the model?	$(1.5 \times 10^8) \div (3 \times 10^6) = 0.5 \times 10^2 = 50$ m
10	How far should the Moon be from Earth in the model? Is this easy to show? Explain.	$(3.84 \times 10^5) \div (3 \times 10^6) = 0.128$ m ≈ 13 cm. The Earth and Moon would be very close together, and both are tiny at this scale.
11	Draw a labeled plan of your hallway model.	Answers vary.

Answer Key – CP1

Part 1: Big & Small

#	Task	Answer
1	How many flu viruses would fit side by side across a red blood cell? Across a human hair?	$(8 \times 10^{-6}) \div (1 \times 10^{-7}) = 8 \times 10^{1} = 80$ viruses; $(8 \times 10^{-5}) \div (1 \times 10^{-7}) = 8 \times 10^{2} = 800$ viruses
2	How many times larger is Earth's diameter than a human hair? (Convert to the same units first: 1 km = 10^{3} m.)	Earth: 1.27×10^{4} km = 1.27×10^{7} m. $(1.27 \times 10^{7}) \div (8 \times 10^{-5}) \approx 0.159 \times 10^{12} \approx 1.59 \times 10^{11}$ times
3	Simplify $(6 \times 10^{-3})(4 \times 10^{8})$. Write the answer in scientific notation.	$24 \times 10^{5} = 2.4 \times 10^{6}$
4	Explain why 24×10^{5} is not in scientific notation, and fix it.	The coefficient 24 is not between 1 and 10. $24 \times 10^{5} = 2.4 \times 10^{6}$.

Part 2: Light & Travel

#	Task	Answer
5	Find the light travel time from the Sun to Earth and from Earth to Mars (closest). Give each in seconds and minutes.	Sun: 500 s ≈ 8.3 min. Mars: $(5.6 \times 10^{7}) \div (3 \times 10^{5}) \approx 1.87 \times 10^{2}$ ≈ 187 s ≈ 3.1 min
6	A light-year is the distance light travels in one year. Use 3.15×10^{7} seconds per year to find a light-year in km.	$(3 \times 10^{5})(3.15 \times 10^{7}) = 9.45 \times 10^{12}$ km
7	How many light-years away is Proxima Centauri?	$(4.0 \times 10^{13}) \div (9.45 \times 10^{12}) \approx 4.2$ light-years
8	A spacecraft travels 6×10^{4} km per hour. How many years would it take to reach Proxima Centauri? (1 year ≈ 8.76×10^{3} hours)	$(4.0 \times 10^{13}) \div (6 \times 10^{4}) \approx 6.67 \times 10^{8}$ hours; $\div (8.76 \times 10^{3}) \approx 7.6 \times 10^{4}$ years (about 76,000 years)

Part 3: Design a Hallway Model

#	Task	Answer
9	Choose a scale (1 m in the model = ? km) so that the Sun-to-Earth distance fits with room to spare. Write the scale in scientific notation.	Example: 1 m = 5×10^{6} km → Sun to Earth = 30 m.
10	Mars averages 2.28×10^{8} km from the Sun. Find the model distances for Earth and Mars.	Example scale: Earth 30 m, Mars $(2.28 \times 10^{8}) \div (5 \times 10^{6}) =$ 45.6 m.
11	At your scale, how big would Earth's diameter be? Could you see it? Explain.	Example scale: $(1.27 \times 10^{4}) \div (5 \times 10^{6}) = 2.54 \times 10^{-3}$ m ≈ 2.5 mm, about the size of a grain of sand.

Answer Key – Honors

Part 1: Powers of Ten

#	Task	Answer
1	Place the flu virus, red blood cell, hair, Earth's diameter, Earth-Sun distance, and Proxima distance on the powers-of-ten line (estimate each power).	Virus 10^{-7}, cell ≈ 10^{-5} (8×10^{-6}), hair ≈ 10^{-4} (8×10^{-5}), Earth ≈ 10^{7}, Earth-Sun ≈ 10^{11} (1.5×10^{11} m), Proxima ≈ 10^{16} (4×10^{16} m, beyond the line's end)
2	Explain why equal spaces on this line represent multiplying by the same number, not adding.	Each step multiplies by 10 (or 100 per tick). Equal distances mean equal ratios.
3	Simplify $(2 \times 10^{3})^{3}$ and $(8 \times 10^{-5})^{2} \div (4 \times 10^{-7})$. Write answers in scientific notation.	8×10^{9}; $(64 \times 10^{-10}) \div (4 \times 10^{-7}) = 16 \times 10^{-3} = 1.6 \times 10^{-2}$
4	A student wrote $(3 \times 10^{4})(5 \times 10^{-9}) = 15 \times 10^{-36}$. Identify both errors and correct the answer.	The exponents should be added, not multiplied, and 15 must be rewritten: $15 \times 10^{-5} = 1.5 \times 10^{-4}$

Part 2: Light & Travel

#	Task	Answer
5	Find a light-year in km. Then express the Proxima Centauri distance in light-years.	9.45×10^{12} km; $(4.0 \times 10^{13}) \div (9.45 \times 10^{12}) \approx 4.23$ light-years
6	Find the volume of a cube-shaped box of side 8×10^{-6} m (about the size of a blood cell). How many such boxes fit in 1 cubic millimeter (10^{-9} m^3)?	$(8 \times 10^{-6})^{3} = 512 \times 10^{-18} = 5.12 \times 10^{-16}$ m^3; $10^{-9} \div (5.12 \times 10^{-16}) \approx 1.95 \times 10^{6}$ boxes
7	Light from Proxima left the star how many seconds ago? Express it two ways: in seconds and in years.	$(4.0 \times 10^{13}) \div (3 \times 10^{5}) \approx 1.33 \times 10^{8}$ s ≈ 4.23 years
8	If a probe could travel at 1% of the speed of light, how many years would it take to reach Proxima? Show how the exponent rules make this quick.	At 1% of light speed the trip takes 100 times as long: $4.23 \times 10^{2} \approx 423$ years.

Part 3: Design a Two-Scale Exhibit

#	Task	Answer
9	Choose a distance scale so Mars (2.28×10^{8} km from the Sun) is at 45 m or less. Find the model distances for Earth and Mars.	Example: 1 m = 5.1×10^{6} km → Mars 44.7 m, Earth 29.4 m.
10	Choose a size scale so Earth's diameter is 10 cm. Find the model size of the Moon (diameter 3.47×10^{3} km) and the Sun (diameter 1.39×10^{6} km).	Scale 10 cm = 1.27×10^{4} km. Moon: $3.47 \times 10^{3} \div 1.27 \times 10^{4} \times 10 \approx 2.7$ cm. Sun: $(1.39 \times 10^{6}) \div (1.27 \times 10^{4}) \times 10 \approx 1{,}094$ cm ≈ 10.9 m.

11	Explain why one scale cannot show both sizes and distances well. Use the ratio of the two scales.	The size scale is about 10^4 times larger than the distance scale; at the size scale, Earth would be about 1.2 km from the Sun.

Scoring note: On the powers-of-ten line, Proxima's distance ($\approx 4 \times 10^{16}$ m) falls beyond 10^{14}; students should note it is off the line, which is a good discussion point.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Model fits the chosen space and every scaled distance is calculated correctly. Goes beyond the minimum with a creative, realistic choice.	Model fits the chosen space and every scaled distance is calculated correctly.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Scale model plan shows the scale, every converted distance in scientific notation, and a labeled diagram. Polished, organized, and visually engaging.	Scale model plan shows the scale, every converted distance in scientific notation, and a labeled diagram. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.